

Module 1: Data Attributes & Proximity Metrics

KDD Pipeline, 4 Attribute Types, Boxplots, Euclidean & Cosine Distance

Module I: Introduction to Data Mining & Data Attributes — 14 Topics Syllabus

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| 1. Introduction to Data Mining & KDD | 2. Relational Databases Architecture |
| 3. Data Warehouses Fundamentals | 4. Transactional Databases |
| 5. Advanced Database Systems & Applications | 6. Data Mining Functionalities (6 Types) |
| 7. Classification of Data Mining Systems | 8. Major Issues & Challenges in Data Mining |
| 9. Data & Dataset Anatomy | 10. Data Objects & 4 Attribute Types |
| 11. Basic Statistical Descriptions of Data | 12. Data Visualization Techniques (5 Paradigms) |
| 13. Measuring Data Similarity (Cosine, Jaccard) | 14. Measuring Data Dissimilarity (L1, L2, Lp, Mixed) |

Topic 1: Introduction to Data Mining & The Knowledge Discovery (KDD) Process

Data Mining is the computational process of discovering hidden, previously unknown, valid, novel, potentially useful, and understandable patterns, correlations, anomalies, and actionable knowledge from massive volumes of structured, semi-structured, and unstructured data repositories.

Often called *Knowledge Discovery in Databases (KDD)*, data mining represents the essential algorithmic step within a broader, highly iterative multi-phase engineering pipeline designed to convert raw operational data into high-value strategic decision models.



FIGURE 1.1: THE END-TO-END KNOWLEDGE DISCOVERY IN DATABASES (KDD) PIPELINE

Comprehensive Analysis of the 7 KDD Phases:

- Data Cleaning:** Industrial sensors, transactional points-of-sale, and web forms produce data corrupted by transmission loss, hardware malfunctions, or transcription mistakes. Cleaning eliminates extreme noise and handles missing attributes via statistical imputation.
- Data Integration:** Merges disparate heterogeneous data sources into a unified repository. Resolves semantic conflicts, entity identification problems (e.g., matching `Cust\_ID` with `Client\_Number`), and detects attribute redundancy through correlation analysis.
- Data Selection:** Dynamically isolates task-relevant attributes and subsets required for a specific business query from the massive data lake or enterprise warehouse.
- Data Transformation:** Normalizes numeric scales ($[0, 1]$ Min-Max or Z-Score), aggregates granular records into summary facts, and constructs concept hierarchies for multi-level pattern mining.

5. **Data Mining (Core Engine):** Employs machine learning, statistical pattern recognition, and database search algorithms (such as Apriori, FP-Growth, Decision Trees, K-Means, Neural Networks) to discover recurring structures, decision boundaries, or clusters.
6. **Pattern Evaluation:** Evaluates discovered candidates against rigorous mathematical interestingness metrics (Support, Confidence, Lift, Information Gain, Silhouette Coefficients) to filter out trivial correlations and avoid statistical overfitting.
7. **Knowledge Presentation:** Uses visual analytics, scatter matrices, parallel coordinate graphs, and interactive decision trees to present discoveries in an interpretable format for human domain experts.

Topics 2 – 5: Data Repositories & Advanced Database Architectures

Data mining algorithms must adapt their indexing, traversal, and caching strategies according to the specific data storage architecture:

Repository Type	Architecture & Data Organization	Data Mining Use Cases & Algorithmic Adaptations
2. Relational Databases (RDBMS)	Organized as structured collections of normalized tables linked by foreign keys, accessed via SQL queries. Compliant with ACID properties (Atomicity, Consistency, Isolation, Durability).	Relational association mining, inductive logic programming (ILP), classification of multi-table customer records without creating expensive flat denormalized views.
3. Data Warehouses	Subject-oriented, integrated, time-variant, non-volatile repositories modeled using multidimensional Data Cubes , Fact tables, and Dimension tables for Online Analytical Processing (OLAP).	On-Line Analytical Mining (OLAM), roll-up/drill-down exploration, discovery-driven data cube navigation using statistical anomaly indicators.
4. Transactional Databases	Flat files or nested tables where each record represents a distinct transaction consisting of a unique identifier ('TID') and a set of items purchased simultaneously ($T \subseteq I$).	Frequent itemset mining, cross-selling analysis, market basket analysis, sequential traversal algorithms (Apriori, FP-Growth, ECLAT).
5. Spatial Databases	Stores multidimensional geometric objects (points, lines, polygons, raster grids) indexed via R-Trees, Quad-trees, or k-d trees with GIS coordinate systems.	Spatial autocorrelation analysis, spatial trend mining, density-based geographic clustering (DBSCAN), trajectory tracking.
5. Time-Series & Temporal	Stores time-stamped numeric sequence values recorded at uniform or non-uniform intervals (stock quotes, IoT sensor telemetry, ECG).	Trend analysis, cyclic/seasonal decomposition, Dynamic Time Warping (DTW), autoregressive ARIMA forecasting, motif discovery.
5. Text & Web Databases	Unstructured text documents, HTML web graphs with hyperlinks, XML trees, server clickstream logs, social media graphs.	TF-IDF vector space modeling, Latent Dirichlet Allocation (LDA), PageRank link analysis, sentiment analysis, web usage mining.

Topic 6: Data Mining Functionalities (What Kinds of Patterns Can Be Mined?)

Data mining tasks are broadly divided into two major complementary paradigms: **Descriptive** (summarizing and discovering structural properties of current data) and **Predictive** (building statistical inference models to forecast future outcomes).

Mining Functionality	Paradigm	Mathematical / Algorithmic Basis	Industrial Example
1. Characterization & Discrimination	Descriptive	Summarizing general features of a target class (Characterization) vs. contrasting it against comparative classes (Discrimination) via Attribute-Oriented Induction.	Summarizing characteristics of top 10% spending customers vs. comparing them against occasional buyers.
2. Frequent Patterns, Associations & Correlations	Descriptive	Discovering itemsets appearing with support $\geq \text{min_sup}$ and confidence $\geq \text{min_conf}$. Generating rules $A \Rightarrow B$ satisfying $\text{Lift}(A, B) > 1$.	$\text{buys}(\text{laptop}) \Rightarrow \text{buys}(\text{mouse}) [2\%, 75\%]$.
3. Classification	Predictive	Supervised learning model constructed from labeled training tuples to predict discrete categorical class labels (Decision Trees, Naïve Bayes, SVM, Neural Networks).	Classifying loan applicants as <i>Safe</i> or <i>Risky</i> based on credit score, salary, and debt.
4. Regression	Predictive	Supervised learning mapping data objects to continuous real-valued numerical variables (Linear Regression, Non-linear curve fitting, Ridge/Lasso).	Predicting next quarter's revenue based on advertising expenditure and historical growth.
5. Cluster Analysis	Descriptive	Unsupervised learning grouping objects into clusters such that intra-cluster similarity is maximized and inter-cluster similarity is minimized (K-Means, Hierarchical, DBSCAN).	Discovering distinct customer market segments without predefined class labels.
6. Outlier Analysis (Anomaly Detection)	Descriptive / Predictive	Identifying data objects that deviate significantly from general distribution patterns (Statistical discordance tests, Distance-based, Density-based LOF).	Credit card fraud detection; sudden spike in foreign transactions at 3 AM.

Topic 7: Classification of Data Mining Systems

Data mining systems are categorized across 4 multidimensional axes:

- 1. By Data Models Mined:** Relational, Transactional, Object-Oriented, Spatial, Temporal, Time-Series, Text, Web, Graph, Multimedia, and Streaming Data Mining systems.
- 2. By Types of Knowledge Discovered:** Association Rule systems, Classification & Prediction engines, Clustering systems, Sequential Pattern miners, Outlier detectors, and Deep Representation embedding systems.
- 3. By Underlying Mining Techniques:** Database-oriented (SQL/OLAP-driven), Machine learning, Statistics, Information retrieval, Neural networks, Genetic algorithms, and Visual analytics.
- 4. By Applications Adapted:** Telecommunications, Healthcare, Banking & Finance, Bioinformatics, E-Commerce, Retail, and Intrusion Detection.

Topic 8: Major Issues & Challenges in Data Mining

5 Major Technical Challenges in Modern Data Mining

1. **Mining Methodology & User Interaction:** Mining complex knowledge in multidimensional space; integrating background domain knowledge; designing expressive mining query languages; intuitive visual representation.
2. **Performance, Scalability & Parallelism:** Processing massive petabyte data streams within strict runtime constraints ($O(n)$ or sub-linear complexity); parallel, distributed, and incremental mining algorithms.
3. **Handling Data Diversity & High Dimensionality:** Dealing with noisy, incomplete data; handling extreme sparsity; mitigating the curse of dimensionality where distance metrics lose discriminative power.
4. **Pattern Evaluation & Overfitting:** Separating genuine causal signals from random noise; addressing the multiple hypothesis testing problem in large candidate spaces.
5. **Social Impacts & Privacy Preservation:** Ensuring algorithmic fairness, differential privacy, GDPR compliance, and avoiding bias in automated credit/hiring algorithms.

Topics 9 – 10: Data Objects & 4 Fundamental Attribute Types

A dataset is a collection of **Data Objects** (also known as samples, records, points, vectors, tuples, observations, or instances). A data object represents an individual entity. Data objects are described by a set of **Attributes** (also called features, dimensions, variables, or fields).

Attribute Class	Mathematical Properties	Permissible Operations	Concrete Examples
1. Nominal (Categorical)	Values are distinct symbols or names without any inherent ordering or magnitude.	Equality (=), Inequality ($\neq$), Mode, Frequency counting. (No mean or median).	`Marital_Status` $\in$ {Single, Married, Divorced}, `Eye_Color`, `Zip_Code`.
2. Binary (Special Nominal)	Nominal attribute with exactly 2 states (0 and 1, or True and False). • Symmetric: Both outcomes equally important (e.g., `Gender` M/F). • Asymmetric: Presence (1) is far more significant/rare than absence (0) (e.g., `Medical_Test` + / -).	Simple Matching Coefficient (SMC) for symmetric; Jaccard Coefficient for asymmetric.	`Covid_Test` $\in$ {0, 1}, `Spam_Flag` $\in$ {0, 1}, `Item_Purchased`.
3. Ordinal	Values have a meaningful sequential order or ranking, but the distance/difference between successive values cannot be measured.	Order comparison ($<$, $\leq$, $>$, $\geq$), Median, Percentiles, Rank correlation (Spearman's ρ).	`Customer_Satisfaction` $\in$ {Poor, Fair, Good, Excellent}, `Academic_Grade` (A, B, C, D, F).
4. Numeric: Interval-Scaled	Measured on a scale of equal-sized units. Meaningful differences between values, but no true zero point (arbitrary zero).	Addition (+), Subtraction (-), Mean, Standard Deviation. (Ratios are meaningless).	`Temperature` in $^{\circ}\text{C}$ or $^{\circ}\text{F}$ (0°C does not mean zero heat; 20°C is not twice as hot as 10°C), `Calendar_Year`.
4. Numeric: Ratio-Scaled	Numeric attribute with an absolute, inherent true zero point ($0.0 = \text{none}$).	All arithmetic operations ($+$, $-$, $\times$, $\div$), Geometric Mean, Harmonic Mean, Coefficient of Variation.	`Salary` (\n0 means zero earnings; \$100k is exactly 2\times\$50k), `Age`, `Weight`, `Length`, `Price`.

Topic 11: Basic Statistical Descriptions of Data

To understand the fundamental characteristics, central tendency, and dispersion of numeric data, we compute descriptive summary statistics.

1. Measures of Central Tendency

- **Arithmetic Mean:** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. Sensitive to extreme outliers.
- **Trimmed Mean:** Computed after discarding a fixed percentage (e.g., 5%) of the lowest and highest values to improve robustness against outliers.

- **Median:** Middle value in sorted order. For even n , average of middle two. Robust measure of central tendency:

$$\text{Median} = \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{x_{n/2} + x_{(n/2)+1}}{2} & \text{if } n \text{ is even} \end{cases}$$

- **Mode:** The most frequently occurring value in the dataset. A dataset can be unimodal, bimodal, multimodal, or have no mode.
- **Midrange:** Average of smallest and largest values: $\text{Midrange} = \frac{\min + \max}{2}$.

Memory Key: Skewness & Mean vs. Median Relationship

Symmetric Distribution: Mean = Median = Mode.

Positively Skewed (Right-Skewed): Tail extends to the right; Mode < Median < Mean.

Negatively Skewed (Left-Skewed): Tail extends to the left; Mean < Median < Mode.

2. Measures of Dispersion & The Five-Number Summary

- **Range:** $\text{Range} = \max(x) - \min(x)$.
- **Quartiles:** Q_1 (25th percentile / First Quartile), Q_2 (50th percentile / Median), Q_3 (75th percentile / Third Quartile).
- **Interquartile Range (IQR):** Distance between 75th and 25th percentiles:

$$\text{IQR} = Q_3 - Q_1$$

- **Five-Number Summary:** $[\min, Q_1, \text{Median}, Q_3, \max]$.
- **Boxplot Outlier Boundary Rule (Tukey's Rule):** Any data point falling beyond the inner fences is flagged as an outlier:

$$\text{Lower Fence} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper Fence} = Q_3 + 1.5 \times \text{IQR}$$

- **Sample Variance (s^2) & Standard Deviation (s):**

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}$$

Step-by-Step Solved Problem: Five-Number Summary & Outlier Detection

Dataset (Sorted): $S = [12, 15, 18, 22, 24, 25, 28, 30, 32, 35, 38, 42, 45, 95]$ ($n = 14$).

Calculations:

1. $\min = 12, \quad \max = 95$.
2. Median (Q_2) = $\frac{x_7 + x_8}{2} = \frac{28 + 30}{2} = \mathbf{29.0}$.
3. Q_1 = Median of lower half $[12, 15, 18, 22, 24, 25, 28] = \mathbf{22.0}$.
4. Q_3 = Median of upper half $[30, 32, 35, 38, 42, 45, 95] = \mathbf{38.0}$.
5. $\text{IQR} = Q_3 - Q_1 = 38.0 - 22.0 = \mathbf{16.0}$.
6. Lower Fence = $22.0 - 1.5(16.0) = 22.0 - 24.0 = \mathbf{-2.0}$.
7. Upper Fence = $38.0 + 1.5(16.0) = 38.0 + 24.0 = \mathbf{62.0}$.
8. **Outlier Identification:** Value $95 > 62.0 \implies \mathbf{95}$ is confirmed as an extreme outlier!

Topic 12: Data Visualization Techniques

Visualization Category	Core Visual Mapping Mechanism	Common Visual Types
1. Pixel-Oriented Techniques	Maps each attribute value to a colored pixel; arranges pixels in recursive space-filling curve windows.	Space-filling curve pixel windows, Circle segments.
2. Geometric Projection	Projects multi-dimensional datasets onto 2D/3D visual subspaces.	Scatter Plot Matrices , Parallel Coordinates (attributes mapped to parallel vertical axes; tuples drawn as polylines).
3. Icon-Based Techniques	Maps attribute dimensions to visual facial or shape features of graphic icons.	Chernoff Faces (eye size, smile curvature, head shape), Stick figures.
4. Hierarchical Techniques	Subdivides attribute space into hierarchical nested subspaces.	Dimensional Stacking, Treemaps (nested rectangular area proportional to value).
5. Visualizing Complex Data	Renders topological networks, text tags, and geographical thematic maps.	Node-link graphs, Chord diagrams, Word clouds, Choropleth maps.

Topics 13 – 14: Measuring Data Similarity and Dissimilarity

Proximity between two objects $x = (x_1, x_2, \dots, x_p)$ and $y = (y_1, y_2, \dots, y_p)$ is quantified using **Distance (Dissimilarity)** $d(x, y)$ or **Similarity** $s(x, y)$.

1. Proximity Measures for Numeric Attributes

Minkowski Distance (L_p Norm):
$$d(x, y) = \left(\sum_{i=1}^p |x_i - y_i|^p \right)^{1/p}$$

Metric Name	Parameter	Mathematical Formula	Properties & Geometric Interpretation
Manhattan Distance (L_1)	$p = 1$	$d(x, y) = \sum_{i=1}^p x_i - y_i $	City-block grid distance. Robust against single-feature extreme outliers.
Euclidean Distance (L_2)	$p = 2$	$d(x, y) = \sqrt{\sum_{i=1}^p (x_i - y_i)^2}$	Standard straight-line Euclidean distance in Cartesian space.
Supremum Distance (L_∞)	$p \rightarrow \infty$	$d(x, y) = \max_{i=1}^p x_i - y_i $	Maximum coordinate difference along any single dimension (Chebyshev distance).

2. Proximity for Asymmetric Binary & Document Vector Data

Metric Name	Mathematical Formulation	Core Use Case
Simple Matching Coeff. (SMC)	$SMC(x, y) = \frac{q + t}{q + r + s + t}$	Symmetric binary data (where 0 – 0 matches t are as important as 1 – 1 matches q).
Jaccard Coefficient (J)	$J(x, y) = \frac{ x \cap y }{ x \cup y } = \frac{q}{q + r + s}, \quad d_J(x, y) = 1 - J(x, y) = \frac{r + s}{q + r + s}$	Asymmetric binary data (ignores joint absence t ; $q = 1 - 1, r = 1 - 0, s = 0 - 1$).
Cosine Similarity	$sim(x, y) = \frac{\vec{x} \cdot \vec{y}}{\ \vec{x}\ \ \vec{y}\ } = \frac{\sum_{i=1}^p x_i y_i}{\sqrt{\sum x_i^2} \sqrt{\sum y_i^2}}$	Text Mining & TF-IDF Vectors: Measures angle θ between vectors; invariant to document length.

Step-by-Step Solved Problem: Proximity Calculations on Vector Data

Given Two Data Points: $x = (3, 0, 4, 1)$, $y = (1, 2, 4, 0)$.

1. **Manhattan Distance (L_1):**

$$d(x, y) = |3 - 1| + |0 - 2| + |4 - 4| + |1 - 0| = 2 + 2 + 0 + 1 = \mathbf{5.0}$$

2. **Euclidean Distance (L_2):**

$$d(x, y) = \sqrt{(3 - 1)^2 + (0 - 2)^2 + (4 - 4)^2 + (1 - 0)^2} = \sqrt{4 + 4 + 0 + 1} = \sqrt{9} = \mathbf{3.0}$$

3. **Supremum Distance (L_∞):**

$$d(x, y) = \max(|3 - 1|, |0 - 2|, |4 - 4|, |1 - 0|) = \max(2, 2, 0, 1) = \mathbf{2.0}$$

4. **Cosine Similarity ($\text{sim}(x, y)$):**

$$\vec{x} \cdot \vec{y} = (3 \times 1) + (0 \times 2) + (4 \times 4) + (1 \times 0) = 3 + 0 + 16 + 0 = 19$$

$$\|\vec{x}\| = \sqrt{3^2 + 0^2 + 4^2 + 1^2} = \sqrt{9 + 0 + 16 + 1} = \sqrt{26} \approx 5.099$$

$$\|\vec{y}\| = \sqrt{1^2 + 2^2 + 4^2 + 0^2} = \sqrt{1 + 4 + 16 + 0} = \sqrt{21} \approx 4.583$$

$$\text{sim}(x, y) = \frac{19}{5.099 \times 4.583} = \frac{19}{23.368} = \mathbf{0.813} \text{ (Cosine Distance} = 1 - 0.813 = \mathbf{0.187})$$

Top BIT Mesra Exam Questions & Answers (Module I)

Q1. Differentiate between OLTP and OLAP systems across 6 distinct technical dimensions. (8 Marks)

Answer: 1. **Primary Purpose:** OLTP supports day-to-day transactional business operations (INSERT/UPDATE); OLAP supports decision support, executive reporting, and analytical queries.

2. **User Base:** OLTP is used by clerks, front-end staff, DBAs; OLAP is used by managers, data scientists, business analysts.

3. **Data Orientation:** OLTP is application-oriented and operational; OLAP is subject-oriented (Sales, Revenue, Customer).

4. **Database Design:** OLTP is highly normalized (3NF/BCNF) to prevent write anomalies; OLAP is denormalized (Star/Snowflake multidimensional schemas).

5. **Access Patterns:** OLTP executes short, atomic read/write transactions; OLAP executes complex read-only aggregate queries spanning millions of rows.

6. **Time Horizon:** OLTP holds current real-time snapshot; OLAP stores multi-year historical time-variant snapshots.

Q2. Why is Jaccard Coefficient preferred over Simple Matching Coefficient for asymmetric binary data? (5 Marks)

Answer: In asymmetric binary datasets (e.g., market basket analysis where thousands of items exist), the absence of an item (0 – 0 match) carries virtually zero predictive significance (e.g., two customers not buying caviar does not indicate similarity). SMC includes negative matches (t) in both numerator and denominator, causing two unrelated customers to appear artificially 99.9% similar. The **Jaccard Coefficient** excludes t entirely, focusing exclusively on joint positive co-occurrences (q), yielding a true measure of affinity.